Akshat Sharma

BS–MS Candidate in Chemical Sciences
Indian Institute of Science Education and Research Kolkata

Email | Website | Google Scholar | ORCID | GitHub

Research Profile

Fourth-year BS-MS student in Chemical Sciences at IISER Kolkata, with a minor in Computational and Data Sciences. My research experience spans molecular self-assembly, electronic-structure analysis, and the development of quantitative frameworks for molecular and statistical complexity. I am particularly interested in statistical mechanics, information theory, soft matter, and theoretical approaches to self-organization and biological physics.

Education

Indian Institute of Science Education and Research Kolkata *Aug 2023–Present*
BS–MS Dual Degree Programme
Major: Chemical Sciences | Minor: Computational and Data Sciences
Expected Graduation: May 2028

Research Interests

Statistical Mechanics; Theoretical Biophysics; Soft Matter; Information Theory; Self-Organization and Complex Systems

Publications & Preprints

Sharma, A. Quantifying the physical complexity of chemical architectures through electron-density topology. *ChemRxiv* (2026). doi:10.26434/chemrxiv.15000864/v4.
Accepted for publication in *Journal of Molecular Modelling*.

Sharma, A. A Unified Discrete and Continuous Theory of Core-Halo Complexity Maximizers. *arXiv:2607.17907* (2026).
Manuscript under review at *Journal of Mathematical Physics*.

Research Experience

Undergraduate Researcher *Dec 2023–Present*
Department of Chemical Sciences, IISER Kolkata
Supervisor: **Prof. Soumyajit Roy**

- Investigate polyoxometalates, soft-oxometalates, and self-assembly phenomena.
- Synthesized Mo_{132} Keplerates and Mo_{154} rings; characterized materials using PXRD, UV–Vis, and Raman spectroscopy.
- Performed photocatalysis and electrocatalysis; phase transition analysis of paracetamol and ibuprofen

Research Intern *May 2025–Jul 2025*
Department of Chemistry, IIT Ropar
Supervisor: **Prof. C. M. Nagaraja**

- Synthesized N-heterocyclic carbene (NHC) ligands and porous organic polymer (POP) precursors for carbon-capture applications.
- Conducted air-sensitive synthesis and monitored reactions under inert conditions.
- Characterized reaction intermediates using NMR spectroscopy.

Independent Theoretical Research

Core–Halo Complexity in Polyelectrolyte Systems *Sep 2026–Present*
Independent theoretical investigation

- Investigate a possible connection between thermodynamic variance extrema and complexity maximization in a confined logarithmic-potential model motivated by Manning condensation.
- Derived asymptotic bounds and analysed the global variance maximum in the single-counterion ($M = 1$) regime within the specified continuous-functional formulation.
- Exploring extensions to interacting many-body systems ($M > 1$), including mean-field Poisson–Boltzmann descriptions.

Information-Theoretic Analysis of a Protein MD Trajectory *Aug 2026*
Independent Computational Project

- Analysed lysozyme molecular-dynamics trajectories using GROMACS, tracking backbone RMSD and Ramachandran dihedral angles.
- Used Python to calculate Rényi-entropy-based descriptors of coarse-grained conformational distributions and visualise them on a probability simplex.

A Unified Discrete and Continuous Theory of Core–Halo Complexity Maximizers *Jun 2026–Present*
Independent theoretical research

- Developed a variational framework for studying generalized statistical complexity in discrete and continuous probability distributions.
- Derived the universal core-halo structure of complexity-maximizing distributions.
- Investigated relationships between entropy-based complexity measures and information geometry.
- Manuscript under review at *Journal of Mathematical Physics*.

Quantum Assembly Index (A_ρ) *Mar 2026–Oct 2026*
Molecular complexity from electron-density topology

- Developed an electron-density-topology-based descriptor for quantifying molecular complexity using QTAIM analysis.
- Formulated the descriptor to connect continuous electronic structure with quantitative measures of molecular complexity.
- Benchmarked the descriptor against selected existing discrete molecular-complexity measures.
- Associated manuscript accepted for publication in *Journal of Molecular Modelling*.

Electronic Structure and Bonding of Zintl Clusters *Jan 2026–Mar 2026*
Independent computational study

- Performed DFT geometry optimizations and electronic-structure calculations for homoatomic and heteroatomic Zintl clusters using ORCA 6.0.
- Analyzed bonding topologies in homoatomic and heteroatomic clusters.

Conferences & Presentations

Poster Presentation (scheduled) *30–31 Oct 2026*
Sharma, A. *A Unified Discrete and Continuous Theory of Core–Halo Complexity Maximizers.*
National Conference on Condensed Matter, Materials Science & Statistical Physics (CMMSSP 2026), Kolkata, India.

Technical Skills

Programming: Python, C; numerical computation and mathematical modelling.
Computational Chemistry: DFT calculations, geometry optimization, vibrational-frequency analysis, electron-density topology; ORCA 6.0, Multiwfn, Avogadro.
Experimental Methods: Polyoxometalate synthesis, air-sensitive synthesis, UV–Vis, Raman spectroscopy, PXRD, SCXRD.
Scientific Software: Olex2, CrysAlisPro, Mercury, Origin, LaTeX.

Selected Coursework

Physics & Chemistry: Statistical Thermodynamics, Phase equilibrium Thermodynamics, Quantum Chemistry
Mathematics, Statistics & Theory: Real Analysis, Linear Algebra, Probability, Statistics, Information Theory, Graph Theory, Group Theory.
Computing: Data Structures and Algorithms, Computer Architecture, Artificial Intelligence for Data Science.

Awards & Recognition

Reliance Foundation Undergraduate Scholarship *2024*
Selected among the top 5,000 undergraduate students in India for academic excellence.

Leadership & Service

Office Bearer, Tennis Club *Aug 2024–May 2025*
Indian Institute of Science Education and Research Kolkata.